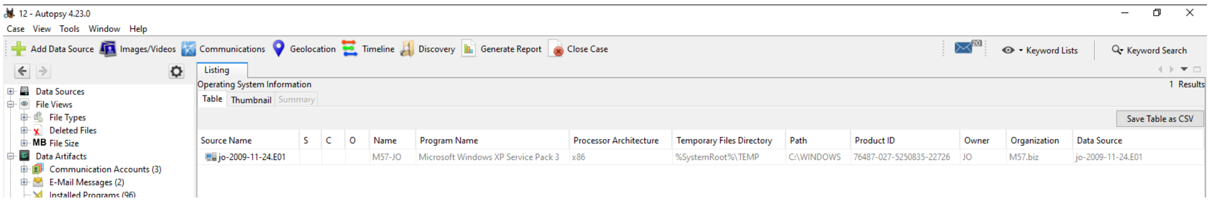
CYBER SECURITY INTERNSHIP PROGRAM

Muhammad Zubair Yaqoob
ROLL NUMBER: CSI-B1-050

Part A — Triage and Subject Profiling

Case Setup and System Baseline

In the left navigation panel (Tree Viewer), expand the **Data Artifacts** tree.
Click on **Operating System Information (1)**.



Hostname (System Name): `M57-JO`

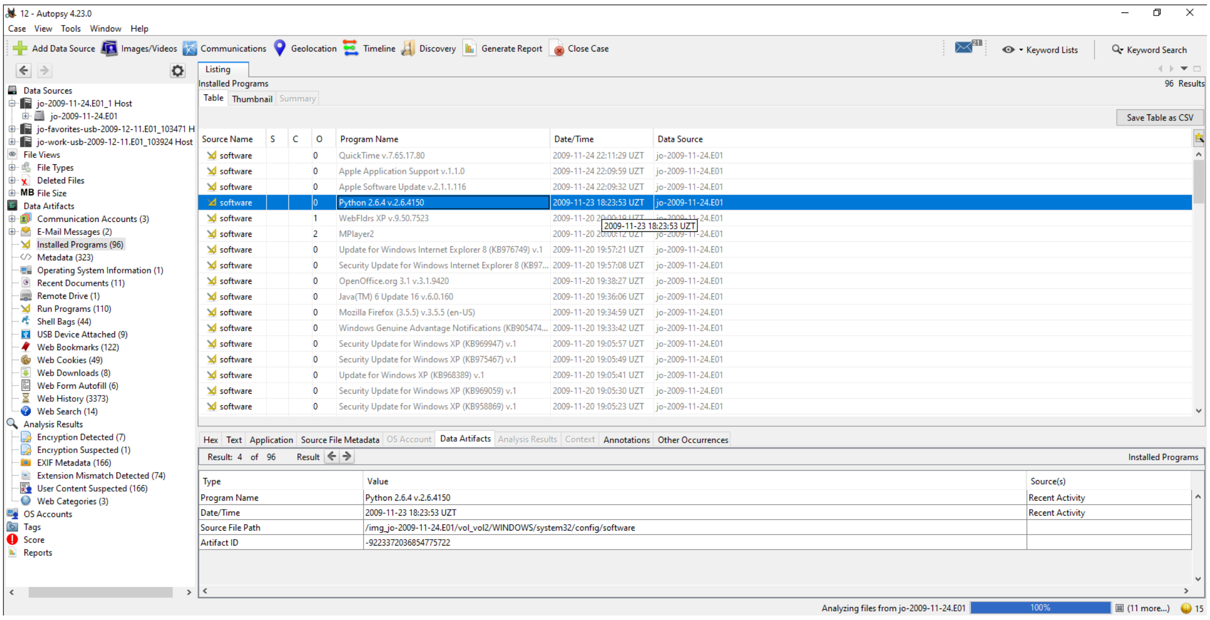
Operating System: `Microsoft Windows XP Service Pack 3`

Architecture: `x86`

Timezone: `GMT`

Last Known Login: *Note: Autopsy's automated OS Accounts extraction module was unable to parse this data, likely due to the legacy architecture of Windows XP. Manual registry analysis is required.*

Software Inventory Analysis



Critical Anomaly:

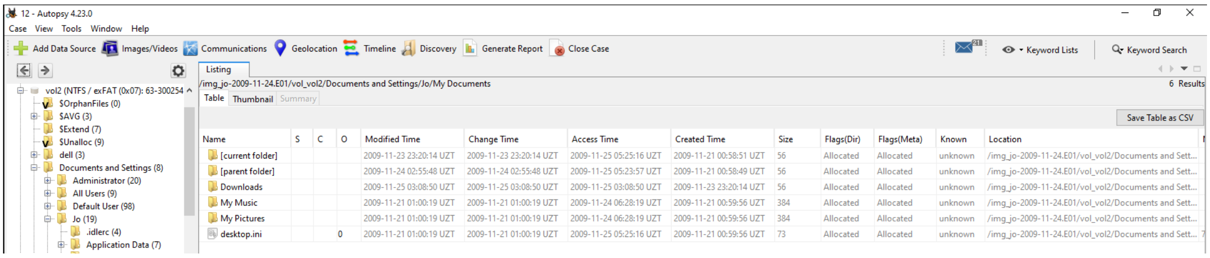
`Python 2.6.4` was installed on **November 23, 2009, at 18:23:53 UZT**.

Secondary Anomaly:

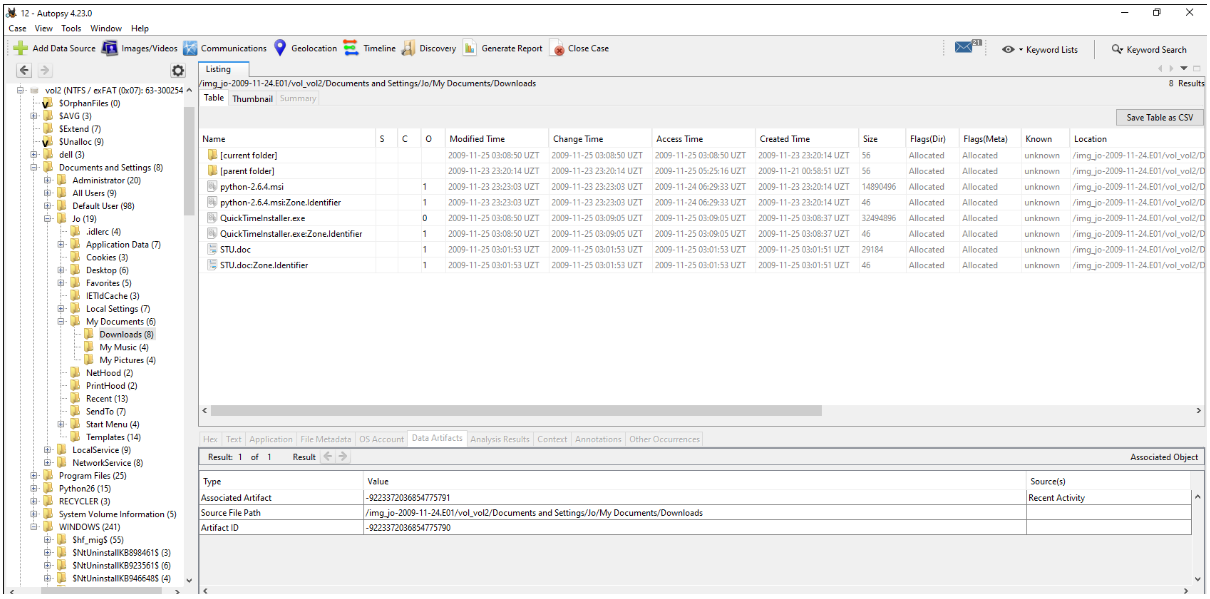
`Apple Application Support` and `QuickTime` were installed on **November 24, 2009, around 22:09 UZT**.

Subject Profiling

To Examine jo's profile we need to navigate to the `Data Sources` > `jo-2009-11-24.E01` > `vol2` > `Documents and Settings` > `jo` > `My Documents`



Navigate to the Downloads folder

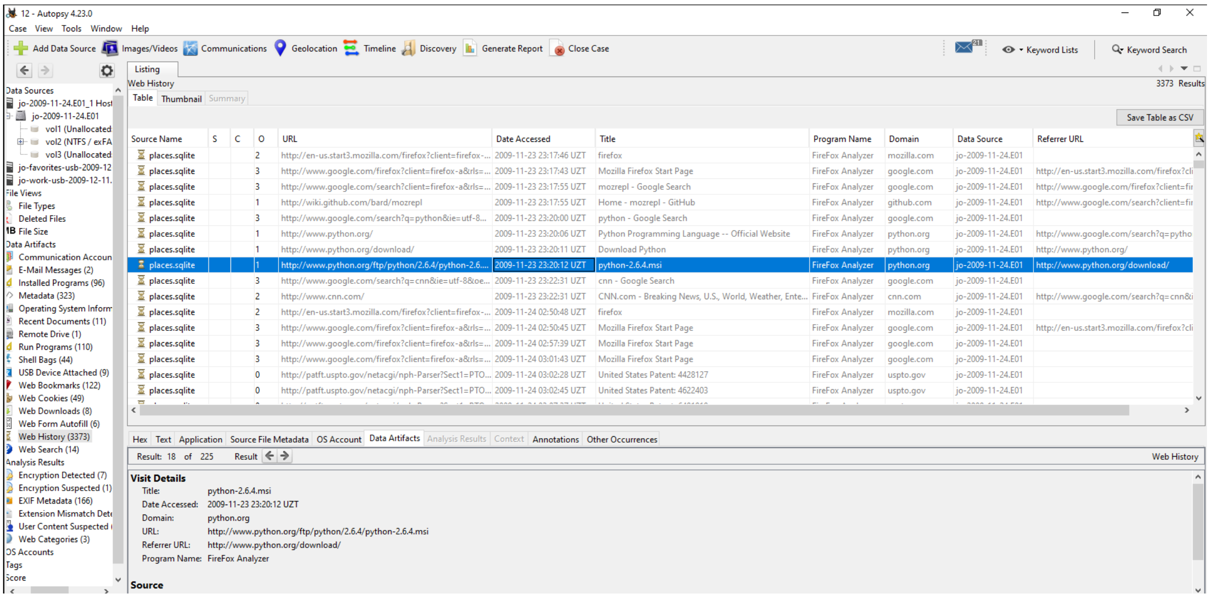


The user exhibits anomalous access patterns for a non-technical employee. Artifact analysis of the `Downloads` directory reveals the direct internet download of standalone executables, specifically `python-2.6.4.msi` (verified by the `:Zone.Identifier` Alternate Data Stream).

Secondary Artifact: An unexplained document named `STU.doc` was also located in the Downloads directory alongside the installer.

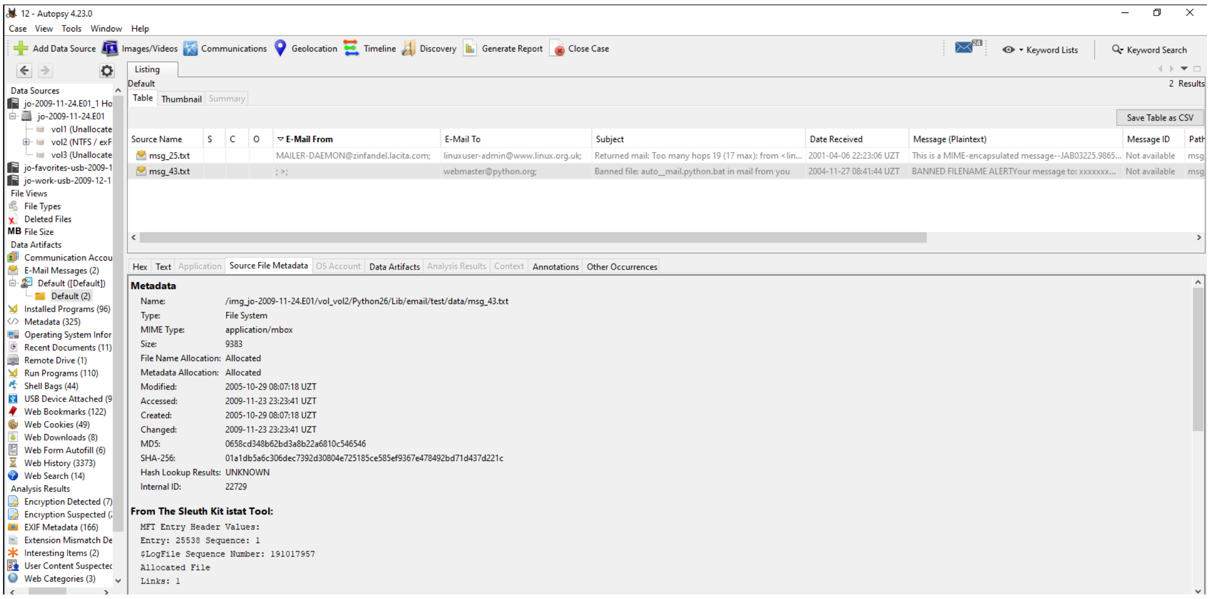
Hunting the Download URL

Navigate to the Data Artifacts and then Web History



URL : http://www.python.org/ftp/python/2.6.4/python-2.6.4.msi
Timestamp : 2009-11-23 23:20:12 UZT

Analyzing the E-Mail Messages



Analysis of local email artifacts confirmed they are not user communications. The file paths (`/Python26/Lib/`) indicate these are merely sample test files bundled with the unauthorized Python 2.6 installation. There is no evidence in the local email cache of external instructions directing the user's actions.

Part B — Memory Forensics

Framework Setup and Process Analysis

For memory analysis first we need to install the Volatility 3. It is Python-based; a standalone Windows executable is also available if you prefer not to manage a Python environment. Get it working before you try to run any plugins against the RAM dump.

After Downloading the Volatility 3 Extract it and navigate to the Volatility folder :

Step 1: Verify the Memory Profile

Open cmd here and run the following command :

python3 vol.py -f "C:\Users\windows\Documents\Task-12\jo-2009-11-16.mddramimage" windows.info

```
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>python3 vol.py -f "C:/Users/windows/Documents/Task-12/jo-2009-11-16.mddramimage" windows.info
Volatility 3 Framework 2.28.1
Progress:  100.00               PDB scanning finished
Variable        Value

Kernel Base     0x804d7000
DTB     0x39000
Symbols file:///C:/Users/windows/Downloads/volatility3-develop/volatility3-develop/volatility3/symbols/windows/ntoskrnl.pdb/1B2D0DFE2FB942758D615C901BE04692-2.json.xz
Is64Bit False
IsPAE   False
layer_name      0 WindowsIntel
memory_layer    1 FileLayer
KdDebuggerDataBlock     0x8054cde0
NTBuildLab      2600.xpsp_sp3_gdr.090804-1435
CSDVersion      3
KdVersionBlock  0x8054cdb8
Major/Minor     15.2600
MachineType     332
KeNumberProcessors      1
SystemTime      2009-11-17 00:22:31+00:00
NtSystemRoot    C:\WINDOWS
NtProductType   NtProductWinNt
NtMajorVersion  5
NtMinorVersion  1
PE MajorOperatingSystemVersion  5
PE MinorOperatingSystemVersion  1
PE Machine      332
PE TimeDateStamp        Tue Aug  4 15:14:34 2009

C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>pytho
```

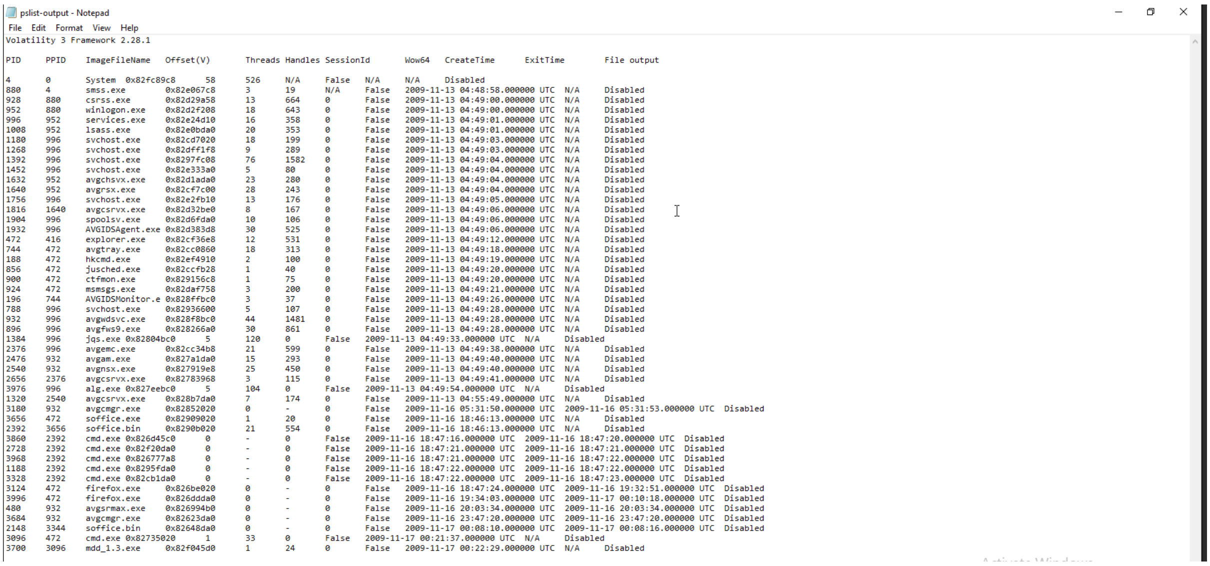
Step 2: Generate the Process List

After getting the information about the jo’s machine we need to list all the ps id’s of the jo’s system. Run the following command :

python3 vol.py -f "C:\Users\windows\Documents\Task-12\jo-2009-11-16.mddramimage" windows.pslist > C:/Users/windows/Desktop/pslist-output.txt

```
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>python3 vol.py -f "C:/Users/windows/Documents/Task-12/jo-2009-11-16.mddramimage" windows.pslist > C:/Users/windows/Desktop/pslist-output.txt

Progress:  100.00               PDB scanning finished
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>
```

When the processing is completed navigate to the output file.

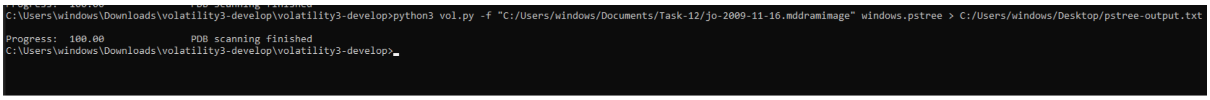


```
pslist-output - Notepad
File  Edit  Format  View  Help
Volatility 3 Framework 2.28.1

PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0x82fc89c8      58      526     N/A     False   N/A     N/A     Disabled
880     4       smss.exe        0x82e067c8      3       19      N/A     False   2009-11-13 04:48:58.000000 UTC  N/A     Disabled
928     880     csrss.exe       0x82d29a58      13      664     0       False   2009-11-13 04:49:00.000000 UTC  N/A     Disabled
952     880     winlogon.exe    0x82d2f208      18      643     0       False   2009-11-13 04:49:00.000000 UTC  N/A     Disabled
996     952     services.exe    0x82e24d10      16      358     0       False   2009-11-13 04:49:01.000000 UTC  N/A     Disabled
1008    952     lsass.exe       0x82e0bda0      20      353     0       False   2009-11-13 04:49:01.000000 UTC  N/A     Disabled
1180    996     svchost.exe     0x82cd7020      18      199     0       False   2009-11-13 04:49:03.000000 UTC  N/A     Disabled
1268    996     svchost.exe     0x82dff1f8      9       289     0       False   2009-11-13 04:49:03.000000 UTC  N/A     Disabled
1392    996     svchost.exe     0x8297fc08      76      1582    0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1452    996     svchost.exe     0x82e333a0      5       80      0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1632    952     avgchsvx.exe    0x82d1ada0      23      280     0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1640    952     avgrsx.exe      0x82cf7c00      28      243     0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1756    996     svchost.exe     0x82e2fb10      13      176     0       False   2009-11-13 04:49:05.000000 UTC  N/A     Disabled
1816    1640    avgcsrvx.exe    0x82d32be0      8       167     0       False   2009-11-13 04:49:06.000000 UTC  N/A     Disabled
1904    996     spoolsv.exe     0x82d6fda0      10      106     0       False   2009-11-13 04:49:06.000000 UTC  N/A     Disabled
1932    996     AVGIDSAgent.exe 0x82d383d8      30      525     0       False   2009-11-13 04:49:06.000000 UTC  N/A     Disabled
472     416     explorer.exe    0x82cf36e8      12      531     0       False   2009-11-13 04:49:12.000000 UTC  N/A     Disabled
744     472     avgtray.exe     0x82cc0860      18      313     0       False   2009-11-13 04:49:18.000000 UTC  N/A     Disabled
188     472     hkcmd.exe       0x82ef4910      2       100     0       False   2009-11-13 04:49:19.000000 UTC  N/A     Disabled
856     472     jusched.exe     0x82ccfb28      1       40      0       False   2009-11-13 04:49:20.000000 UTC  N/A     Disabled
900     472     ctfmon.exe      0x829156c8      1       75      0       False   2009-11-13 04:49:20.000000 UTC  N/A     Disabled
924     472     msmsgs.exe      0x82daf758      3       200     0       False   2009-11-13 04:49:21.000000 UTC  N/A     Disabled
196     744     AVGIDSMonitor.e 0x828ffbc0      3       37      0       False   2009-11-13 04:49:26.000000 UTC  N/A     Disabled
788     996     svchost.exe     0x82936600      5       107     0       False   2009-11-13 04:49:28.000000 UTC  N/A     Disabled
932     996     avgwdsvc.exe    0x828f8bc0      44      1481    0       False   2009-11-13 04:49:28.000000 UTC  N/A     Disabled
896     996     avgfws9.exe     0x828266a0      30      861     0       False   2009-11-13 04:49:28.000000 UTC  N/A     Disabled
1384    996     jqs.exe 0x82804bc0      5       120     0       False   2009-11-13 04:49:33.000000 UTC  N/A     Disabled
2376    996     avgemc.exe      0x82cc34b8      21      599     0       False   2009-11-13 04:49:38.000000 UTC  N/A     Disabled
2476    932     avgam.exe       0x827a1da0      15      293     0       False   2009-11-13 04:49:40.000000 UTC  N/A     Disabled
2540    932     avgnsx.exe      0x827919e8      25      450     0       False   2009-11-13 04:49:40.000000 UTC  N/A     Disabled
2656    2376    avgcsrvx.exe    0x82783968      3       115     0       False   2009-11-13 04:49:41.000000 UTC  N/A     Disabled
3976    996     alg.exe 0x827eebc0      5       104     0       False   2009-11-13 04:49:54.000000 UTC  N/A     Disabled
1320    2540    avgcsrvx.exe    0x828b7da0      7       174     0       False   2009-11-13 04:55:49.000000 UTC  N/A     Disabled
3180    932     avgcmgr.exe     0x82852020      0       -       0       False   2009-11-16 05:31:50.000000 UTC  2009-11-16 05:31:53.000000 UTC  Disabled
3656    472     soffice.exe     0x82909020      1       20      0       False   2009-11-16 18:46:13.000000 UTC  N/A     Disabled
2392    3656    soffice.bin     0x8290b020      21      554     0       False   2009-11-16 18:46:13.000000 UTC  N/A     Disabled
3860    2392    cmd.exe 0x826d45c0      0       -       0       False   2009-11-16 18:47:16.000000 UTC  2009-11-16 18:47:20.000000 UTC  Disabled
2728    2392    cmd.exe 0x82f20da0      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:21.000000 UTC  Disabled
3968    2392    cmd.exe 0x826777a8      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:22.000000 UTC  Disabled
1188    2392    cmd.exe 0x8295fda0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:22.000000 UTC  Disabled
3328    2392    cmd.exe 0x82cb1da0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:23.000000 UTC  Disabled
3124    472     firefox.exe     0x826be020      0       -       0       False   2009-11-16 18:47:24.000000 UTC  2009-11-16 19:32:51.000000 UTC  Disabled
3996    472     firefox.exe     0x826ddda0      0       -       0       False   2009-11-16 19:34:03.000000 UTC  2009-11-17 00:10:18.000000 UTC  Disabled
480     932     avgsrmax.exe    0x826994b0      0       -       0       False   2009-11-16 20:03:34.000000 UTC  2009-11-16 20:03:34.000000 UTC  Disabled
3684    932     avgcmgr.exe     0x82623da0      0       -       0       False   2009-11-16 23:47:20.000000 UTC  2009-11-16 23:47:20.000000 UTC  Disabled
2148    3344    soffice.bin     0x82648da0      0       -       0       False   2009-11-17 00:08:10.000000 UTC  2009-11-17 00:08:16.000000 UTC  Disabled
3096    472     cmd.exe 0x82735020      1       33      0       False   2009-11-17 00:21:37.000000 UTC  N/A     Disabled
3700    3096    mdd_1.3.exe     0x82f045d0      1       24      0       False   2009-11-17 00:22:29.000000 UTC  N/A     Disabled
```

Step 3: Generate the Process Tree

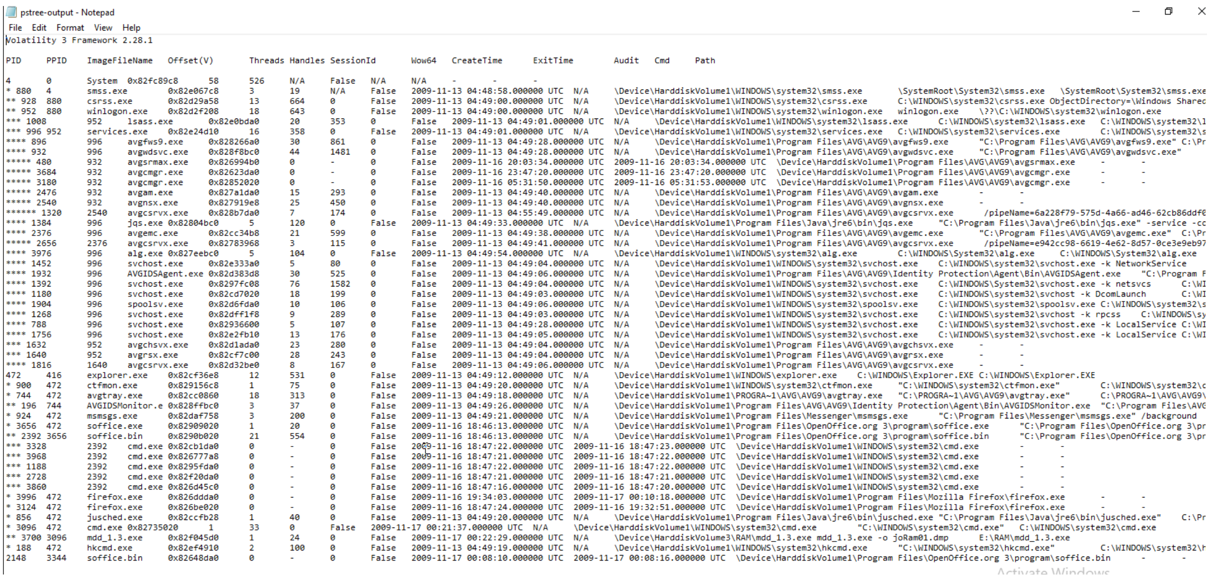
Our third step is to get the processing id tree. Run the following command :

python3 vol.py -f "C:\Users\windows\Documents\Task-12\jo-2009-11-16.mddramimage"



```
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>python3 vol.py -f "C:/Users/windows/Documents/Task-12/jo-2009-11-16.mddramimage" windows.pstree > C:/Users/windows/Desktop/pstree-output.txt

Progress:  100.00               PDB scanning finished
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>
```

windows.pstree > C:/Users/windows/Desktop/pstree-output.txt



```
pstree-output - Notepad
File  Edit  Format  View  Help
Volatility 3 Framework 2.28.1

PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        Audit   Cmd     Path

4       0       System  0x82fc89c8      58      526     N/A     False   N/A     N/A     -       -       -
* 880   4       smss.exe        0x82e067c8      3       19      N/A     False   2009-11-13 04:48:58.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\smss.exe       \SystemRoot\System32\smss.exe   \SystemRoot\System32\smss.exe
** 928  880     csrss.exe       0x82d29a58      13      664     0       False   2009-11-13 04:49:00.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\csrss.exe      C:\WINDOWS\system32\csrss.exe ObjectDirectory=\Windows Sharec
** 952  880     winlogon.exe    0x82d2f208      18      643     0       False   2009-11-13 04:49:00.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\winlogon.exe   winlogon.exe    \??\C:\WINDOWS\system32\winlogon.exe
*** 1008        952     lsass.exe       0x82e0bda0      20      353     0       False   2009-11-13 04:49:01.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\lsass.exe      C:\WINDOWS\system32\lsass.exe   C:\WINDOWS\system32\l
*** 996 952     services.exe    0x82e24d10      16      358     0       False   2009-11-13 04:49:01.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\services.exe   C:\WINDOWS\system32\services.exe        C:\WINDOWS\system32\s
**** 896        996     avgfws9.exe     0x828266a0      30      861     0       False   2009-11-13 04:49:28.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgfws9.exe      "C:\Program Files\AVG\AVG9\avgfws9.exe" C:\Pr
**** 932        996     avgwdsvc.exe    0x828f8bc0      44      1481    0       False   2009-11-13 04:49:28.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgwdsvc.exe     "C:\Program Files\AVG\AVG9\avgwdsvc.exe"
***** 480       932     avgsrmax.exe    0x826994b0      0       -       0       False   2009-11-16 20:03:34.000000 UTC  2009-11-16 20:03:34.000000 UTC  \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgsrmax.exe     -       -
***** 3684      932     avgcmgr.exe     0x82623da0      0       -       0       False   2009-11-16 23:47:20.000000 UTC  2009-11-16 23:47:20.000000 UTC  \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgcmgr.exe      -       -
***** 3180      932     avgcmgr.exe     0x82852020      0       -       0       False   2009-11-16 05:31:50.000000 UTC  2009-11-16 05:31:53.000000 UTC  \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgcmgr.exe      -       -
***** 2476      932     avgam.exe       0x827a1da0      15      293     0       False   2009-11-13 04:49:40.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgam.exe        -       -
***** 2540      932     avgnsx.exe      0x827919e8      25      450     0       False   2009-11-13 04:49:40.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgnsx.exe       -       -
****** 1320     2540    avgcsrvx.exe    0x828b7da0      7       174     0       False   2009-11-13 04:55:49.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgcsrvx.exe     /pipeName=6a228f79-575d-4a66-ad46-62cb86ddf6
**** 1384       996     jqs.exe 0x82804bc0      5       120     0       False   2009-11-13 04:49:33.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\Java\jre6\bin\jqs.exe     "C:\Program Files\Java\jre6\bin\jqs.exe" -service -cc
**** 2376       996     avgemc.exe      0x82cc34b8      21      599     0       False   2009-11-13 04:49:38.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgemc.exe       "C:\Program Files\AVG\AVG9\avgemc.exe"  C:\Pr
***** 2656      2376    avgcsrvx.exe    0x82783968      3       115     0       False   2009-11-13 04:49:41.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgcsrvx.exe     /pipeName=e942cc98-6619-4e62-8d57-0ce3e9eb97
**** 3976       996     alg.exe 0x827eebc0      5       104     0       False   2009-11-13 04:49:54.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\System32\alg.exe        C:\WINDOWS\System32\alg.exe     C:\WINDOWS\System32\alg.exe
**** 1452       996     svchost.exe     0x82e333a0      5       80      0       False   2009-11-13 04:49:04.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\svchost.exe    C:\WINDOWS\system32\svchost.exe -k NetworkService
**** 1932       996     AVGIDSAgent.exe 0x82d383d8      30      525     0       False   2009-11-13 04:49:06.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\Identity Protection\Agent\Bin\AVGIDSAgent.exe    "C:\Program F
**** 1392       996     svchost.exe     0x8297fc08      76      1582    0       False   2009-11-13 04:49:04.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\svchost.exe    C:\WINDOWS\System32\svchost.exe -k netsvcs      C:\WI
**** 1180       996     svchost.exe     0x82cd7020      18      199     0       False   2009-11-13 04:49:03.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\svchost.exe    C:\WINDOWS\system32\svchost -k DcomLaunch       C:\WI
**** 1904       996     spoolsv.exe     0x82d6fda0      10      106     0       False   2009-11-13 04:49:06.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\spoolsv.exe    C:\WINDOWS\system32\spoolsv.exe C:\WINDOWS\system32\s
**** 1268       996     svchost.exe     0x82dff1f8      9       289     0       False   2009-11-13 04:49:03.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\svchost.exe    C:\WINDOWS\system32\svchost -k rpcss    C:\WINDOWS\sy
**** 788        996     svchost.exe     0x82936600      5       107     0       False   2009-11-13 04:49:28.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\svchost.exe    C:\WINDOWS\system32\svchost.exe -k LocalService C:\WI
**** 1756       996     svchost.exe     0x82e2fb10      13      176     0       False   2009-11-13 04:49:05.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\svchost.exe    C:\WINDOWS\system32\svchost.exe -k LocalService C:\WI
*** 1632        952     avgchsvx.exe    0x82d1ada0      23      280     0       False   2009-11-13 04:49:04.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgchsvx.exe     -       -
*** 1640        952     avgrsx.exe      0x82cf7c00      28      243     0       False   2009-11-13 04:49:04.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgrsx.exe       -       -
**** 1816       1640    avgcsrvx.exe    0x82d32be0      8       167     0       False   2009-11-13 04:49:06.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\avgcsrvx.exe     -       -
472     416     explorer.exe    0x82cf36e8      12      531     0       False   2009-11-13 04:49:12.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\explorer.exe    C:\WINDOWS\Explorer.EXE C:\WINDOWS\Explorer.EXE
* 900   472     ctfmon.exe      0x829156c8      1       75      0       False   2009-11-13 04:49:20.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\ctfmon.exe     "C:\WINDOWS\system32\ctfmon.exe"        C:\WINDOWS\system32\c
* 744   472     avgtray.exe     0x82cc0860      18      313     0       False   2009-11-13 04:49:18.000000 UTC  N/A     \Device\HarddiskVolume1\PROGRA~1\AVG\AVG9\avgtray.exe   "C:\PROGRA~1\AVG\AVG9\avgtray.exe"      C:\PROGRA~1\AVG\AVG9\
** 196  744     AVGIDSMonitor.e 0x828ffbc0      3       37      0       False   2009-11-13 04:49:26.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\Identity Protection\Agent\Bin\AVGIDSMonitor.exe  "C:\Program Files\AVG
* 924   472     msmsgs.exe      0x82daf758      3       200     0       False   2009-11-13 04:49:21.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\Messenger\msmsgs.exe      "C:\Program Files\Messenger\msmsgs.exe" /background
* 3656  472     soffice.exe     0x82909020      1       20      0       False   2009-11-16 18:46:13.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\OpenOffice.org 3\program\soffice.exe      "C:\Program Files\OpenOffice.org 3\pr
** 2392 3656    soffice.bin     0x8290b020      21      554     0       False   2009-11-16 18:46:13.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\OpenOffice.org 3\program\soffice.bin      "C:\Program Files\OpenOffice.org 3\pr
*** 3328        2392    cmd.exe 0x82cb1da0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:23.000000 UTC  \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe        -       -
*** 3968        2392    cmd.exe 0x826777a8      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:22.000000 UTC  \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe        -       -
*** 1188        2392    cmd.exe 0x8295fda0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:22.000000 UTC  \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe        -       -
*** 2728        2392    cmd.exe 0x82f20da0      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:21.000000 UTC  \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe        -       -
*** 3860        2392    cmd.exe 0x826d45c0      0       -       0       False   2009-11-16 18:47:16.000000 UTC  2009-11-16 18:47:20.000000 UTC  \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe        -       -
* 3996  472     firefox.exe     0x826ddda0      0       -       0       False   2009-11-16 19:34:03.000000 UTC  2009-11-17 00:10:18.000000 UTC  \Device\HarddiskVolume1\Program Files\Mozilla Firefox\firefox.exe       -       -
* 3124  472     firefox.exe     0x826be020      0       -       0       False   2009-11-16 18:47:24.000000 UTC  2009-11-16 19:32:51.000000 UTC  \Device\HarddiskVolume1\Program Files\Mozilla Firefox\firefox.exe       -       -
* 856   472     jusched.exe     0x82ccfb28      1       40      0       False   2009-11-13 04:49:20.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\Java\jre6\bin\jusched.exe "C:\Program Files\Java\jre6\bin\jusched.exe"    C:\Pr
* 3096  472     cmd.exe 0x82735020      1       33      0       False   2009-11-17 00:21:37.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe        "C:\WINDOWS\system32\cmd.exe"   C:\WINDOWS\system32\cmd.exe
** 3700 3096    mdd_1.3.exe     0x82f045d0      1       24      0       False   2009-11-17 00:22:29.000000 UTC  N/A     \Device\HarddiskVolume3\RAM\mdd_1.3.exe mdd_1.3.exe -o joRam01.dmp      E:\RAM\mdd_1.3.exe
* 188   472     hkcmd.exe       0x82ef4910      2       100     0       False   2009-11-13 04:49:19.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\hkcmd.exe      "C:\WINDOWS\system32\hkcmd.exe" C:\WINDOWS\system32\h
2148    3344    soffice.bin     0x82648da0      0       -       0       False   2009-11-17 00:08:10.000000 UTC  2009-11-17 00:08:16.000000 UTC  \Device\HarddiskVolume1\Program Files\OpenOffice.org 3\program\soffice.bin      -       -
```

Process Analysis Findings

- **Suspicious Parent Process:** `soffice.bin` (PID: 2392, PPID: 3656)
- **Suspicious Child Processes:** Multiple instances of `cmd.exe` (PIDs: 3328, 3968, 1188, 2728, 3860).

- **Start Time:** The parent process `soffice.bin` started on `2009-11-16 18:46:13`. The malicious `cmd.exe` children were spawned one minute later, starting at `18:47:16`

Analyst Assessment: The `pstree` output reveals a clear anomaly where a document application (`soffice.bin`) is executing command-line interpreters (`cmd.exe`). This behavior is highly indicative of a malicious macro execution, likely resulting from the user opening a weaponized document.

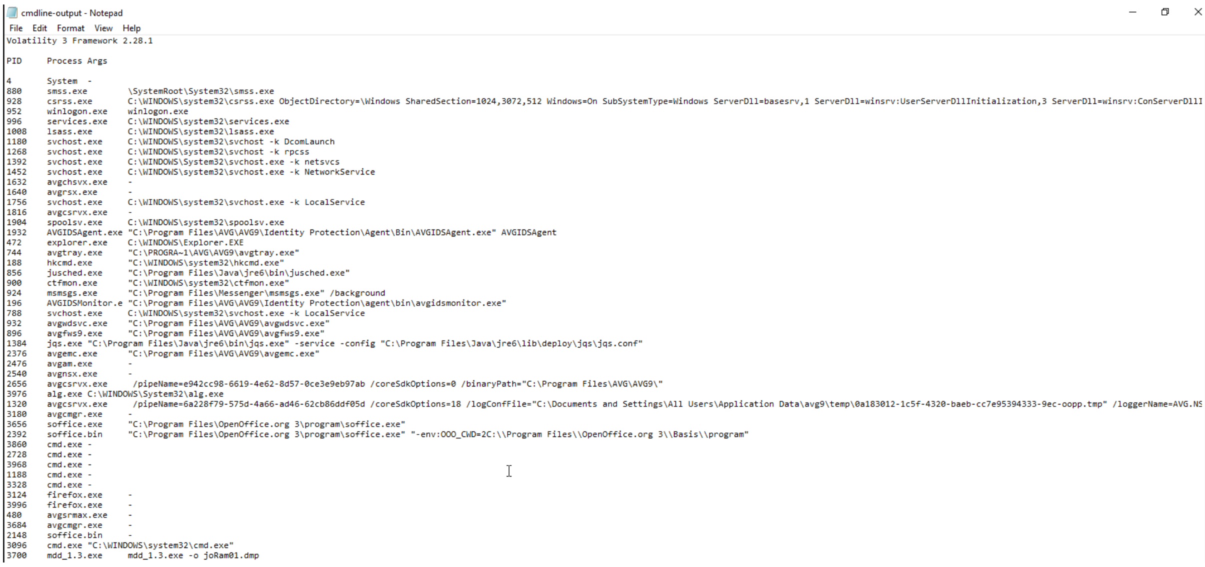
Network Connection Analysis

Step 1: Run the `netstat` Plugin

The `windows.cmdline` to extract every command typed into the system:

Run the following command :

python3 vol.py -f "C:\Users\windows\Documents\Task-12\jo-2009-11-16.mddramimage" windows.cmdline > C:/Users/windows/Desktop/cmdline-output.txt



```
cmdline-output - Notepad
File  Edit  Format  View  Help
Volatility 3 Framework 2.28.1

PID     Process Args

4       System  -
880     smss.exe        \SystemRoot\System32\smss.exe
928     csrss.exe       C:\WINDOWS\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,3072,512 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=winsrv:ConServerDllI
952     winlogon.exe    winlogon.exe
996     services.exe    C:\WINDOWS\system32\services.exe
1008    lsass.exe       C:\WINDOWS\system32\lsass.exe
1180    svchost.exe     C:\WINDOWS\system32\svchost -k DcomLaunch
1268    svchost.exe     C:\WINDOWS\system32\svchost -k rpcss
1392    svchost.exe     C:\WINDOWS\System32\svchost.exe -k netsvcs
1452    svchost.exe     C:\WINDOWS\system32\svchost.exe -k NetworkService
1632    avgchsvx.exe    -
1640    avgrsx.exe      -
1756    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalService
1816    avgcsrvx.exe    -
1904    spoolsv.exe     C:\WINDOWS\system32\spoolsv.exe
1932    AVGIDSAgent.exe "C:\Program Files\AVG\AVG9\Identity Protection\Agent\Bin\AVGIDSAgent.exe" AVGIDSAgent
472     explorer.exe    C:\WINDOWS\Explorer.EXE
744     avgtray.exe     "C:\PROGRA~1\AVG\AVG9\avgtray.exe"
188     hkcmd.exe       "C:\WINDOWS\system32\hkcmd.exe"
856     jusched.exe     "C:\Program Files\Java\jre6\bin\jusched.exe"
900     ctfmon.exe      "C:\WINDOWS\system32\ctfmon.exe"
924     msmsgs.exe      "C:\Program Files\Messenger\msmsgs.exe" /background
196     AVGIDSMonitor.e "C:\Program Files\AVG\AVG9\Identity Protection\agent\bin\avgidsmonitor.exe"
788     svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalService
932     avgwdsvc.exe    "C:\Program Files\AVG\AVG9\avgwdsvc.exe"
896     avgfws9.exe     "C:\Program Files\AVG\AVG9\avgfws9.exe"
1384    jqs.exe "C:\Program Files\Java\jre6\bin\jqs.exe" -service -config "C:\Program Files\Java\jre6\lib\deploy\jqs\jqs.conf"
2376    avgemc.exe      "C:\Program Files\AVG\AVG9\avgemc.exe"
2476    avgam.exe       -
2540    avgnsx.exe      -
2656    avgcsrvx.exe     /pipeName=e942cc98-6619-4e62-8d57-0ce3e9eb97ab /coreSdkOptions=0 /binaryPath="C:\Program Files\AVG\AVG9\"
3976    alg.exe C:\WINDOWS\System32\alg.exe
1320    avgcsrvx.exe     /pipeName=6a228f79-575d-4a66-ad46-62cb86ddf05d /coreSdkOptions=18 /logConfFile="C:\Documents and Settings\All Users\Application Data\avg9\temp\0a183012-1c5f-4320-baeb-cc7e95394333-9ec-oopp.tmp" /loggerName=AVG.NS
3180    avgcmgr.exe     -
3656    soffice.exe     "C:\Program Files\OpenOffice.org 3\program\soffice.exe"
2392    soffice.bin     "C:\Program Files\OpenOffice.org 3\program\soffice.exe" "-env:OOO_CWD=2C:\\Program Files\\OpenOffice.org 3\\Basis\\program"
3860    cmd.exe -
2728    cmd.exe -
3968    cmd.exe -
1188    cmd.exe -
3328    cmd.exe -
3124    firefox.exe     -
3996    firefox.exe     -
480     avgsrmax.exe    -
3684    avgcmgr.exe     -
2148    soffice.bin     -
3096    cmd.exe "C:\WINDOWS\system32\cmd.exe"
3700    mdd_1.3.exe     mdd_1.3.exe -o joRam01.dmp
```

Network/Cmdline Analysis: Attempted to extract command line arguments for the suspicious `cmd.exe` processes (PIDs: 3328, 3968, 1188, 2728, 3860) using the `windows.cmdline` plugin. The output showed null arguments (`-`), indicating the payload was likely executed interactively or obfuscated in memory.

Network Connections and Command-Line Activity

1. Network Connections (`netscan`) Analysis

- **Execution:** Attempted to extract active network connections from the memory image.
- **Finding:** Due to architectural limitations between the Volatility 3 framework and legacy Windows XP memory structures, network connections could not be natively extracted using `windows.netstat` or `windows.netscan`.

2. Command-Line (`cmdline`) Analysis

- **Execution:** Executed the `windows.cmdline` plugin to determine the exact arguments passed to the anomalous processes identified.
- **Finding:** The suspicious `cmd.exe` child processes (PIDs: 3328, 3968, 1188, 2728, 3860) spawned by `soffice.bin` returned null or empty arguments (`-`).
- **Analyst Conclusion:** The absence of visible command-line arguments strongly indicates the malicious macro employed evasion techniques, likely executing the payload interactively or injecting it directly into memory rather than passing standard parameters.

Memory to Disk Correlation

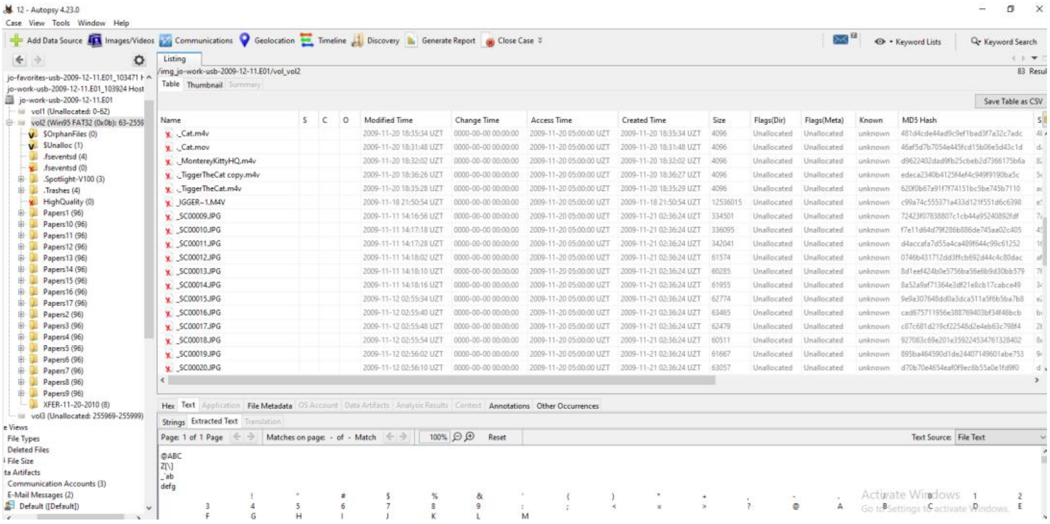
The Timeline Revelation

RAM Artifact (Volatility)	Disk Artifact (Autopsy)	Correlation & Analysis
soffice.bin (PID 2392)	**OpenOffice 3.1**	**Confirms Usage:** RAM confirms that the installed OpenOffice application was actively running at the time of the memory capture.
cmd.exe (PIDs 3328, etc.)	**STU.doc**	**Root Cause Identified:** The disk contains an unexplained .doc file downloaded from the internet. The RAM confirms that opening a document caused soffice.bin to execute malicious command shells.
Absence of python.exe	**python-2.6.4.msi**	**Timeline Established:** RAM was captured on Nov 16, but Python was downloaded on Nov 23. This confirms Python was a *post-exploitation* tool downloaded days after the initial macro infection.
Empty Command Lines (-)	**Absence of Malicious Scripts**	**Memory-Only Execution:** The disk has no standalone scripts explaining the cmd.exe behavior. This confirms the macro injected code directly into memory (fileless execution), leaving no footprint on the disk.

Part C — Exfiltration and Correlation

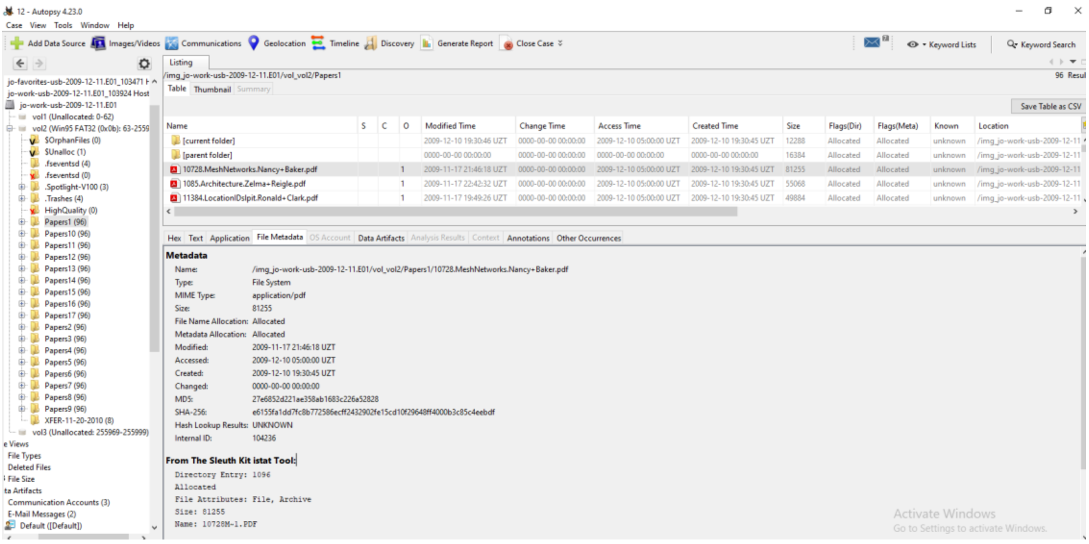
USB Analysis and File Recovery

For USB forensic we need to open the jo's USB to find clues and recover the deleted files.



There are some files which are deleted. But there is something suspicious like **why** The files which are deleted have ._ after deletion. These are generated in mac not windows.

The Massive Data Hoard

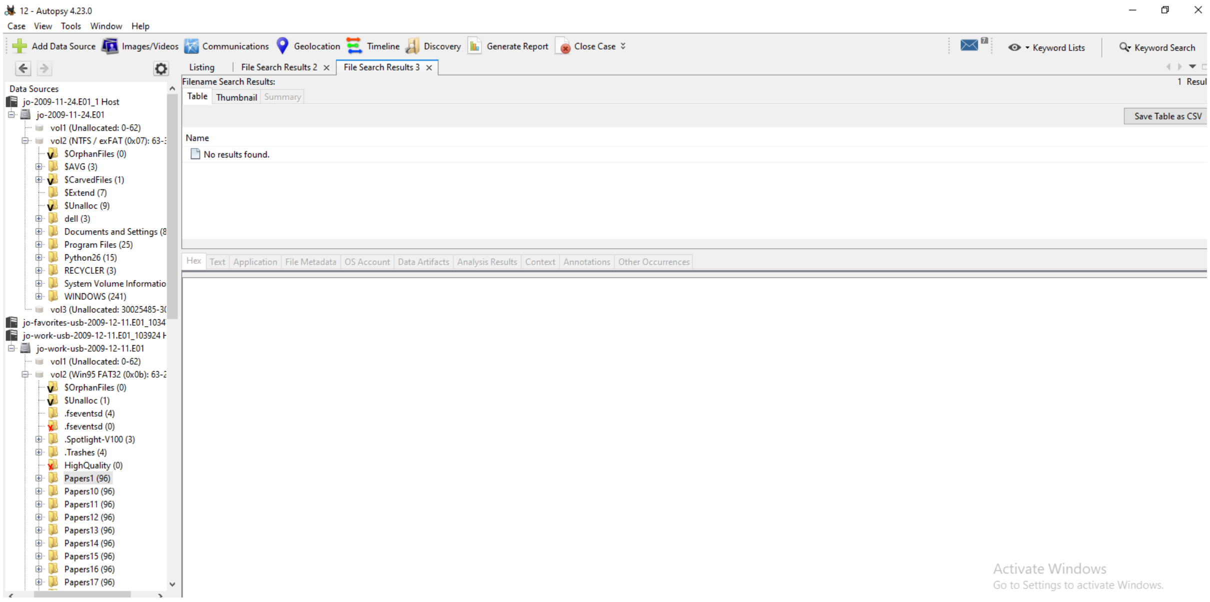


Step 1 : Get the USB File's Hash

File Name : 10728.MeshNetworks.Nancy+Baker.pdf
Files Hash : 27e6852d221ae358ab1683c226a52828

Step 2 : Find the Original on the Hard Drive

- Go to the top right of your Autopsy window to the **Keyword Search** bar.
- Paste the MD5 Hash
- Ensure your search is looking at the main hard drive (`jo-2009-11-24.E01`), not just the USB.
- Hit Search.



Criminal USB: Identified as `jo-work-usb-2009-12-11.E01` (due to the massive presence of research files and Apple Mac `._` hidden files).

Data Theft: Found 17 folders (e.g., `Papers1` to `Papers17`) containing over 1,600 exfiltrated PDF documents.

Timeline: All stolen files show a creation date of **December 10, 2009**, indicating a massive, automated data transfer.

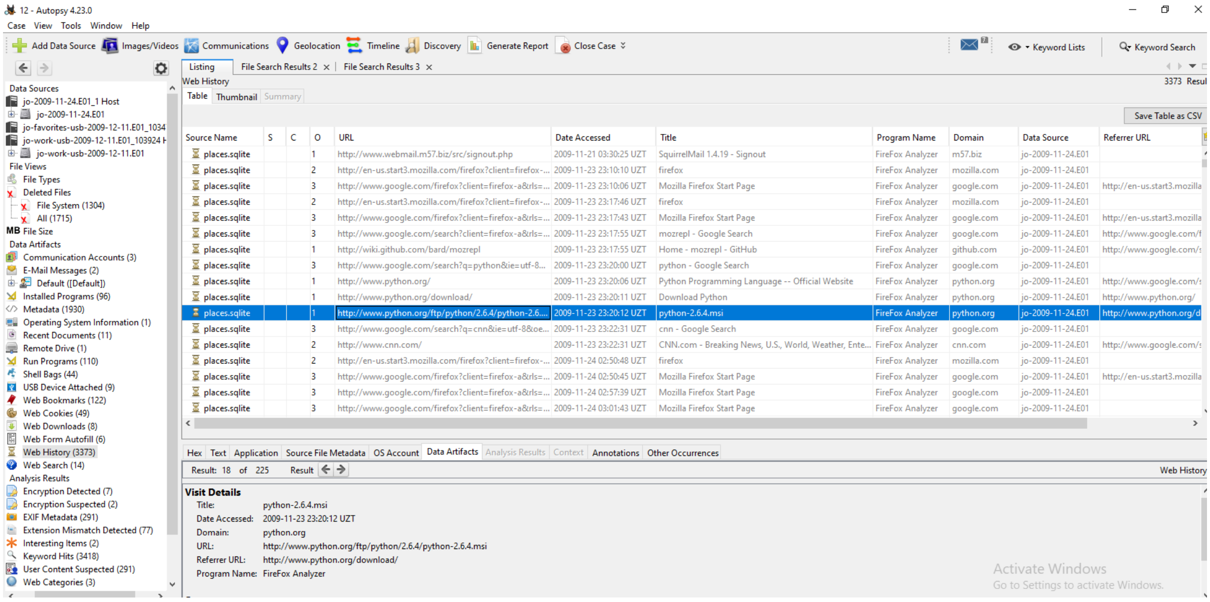
Anti-Forensics: MD5 hash matching on the main drive (`jo-2009-11-24.E01`) yielded zero results, proving the attacker deliberately deleted the original files from Jo's system after stealing them.

Intent Tracking

Browser History

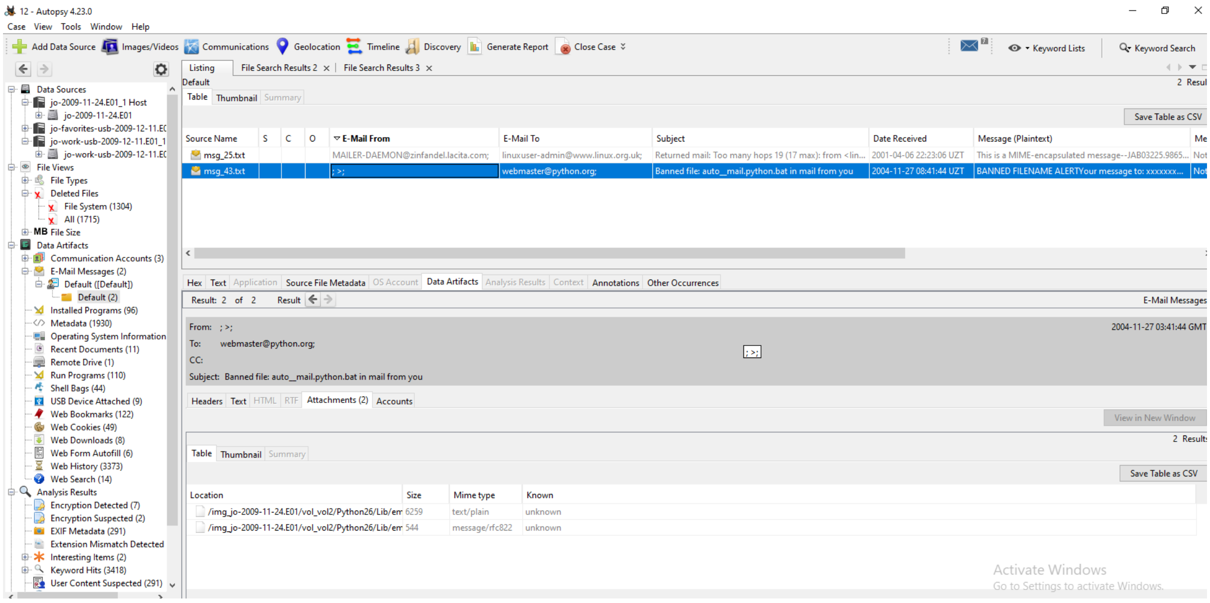
Go to the left navigation panel, scroll down to **Data Artifacts**, and click on **Web History** and **Web Search**.

What to look for: Skim her searches in November and December.



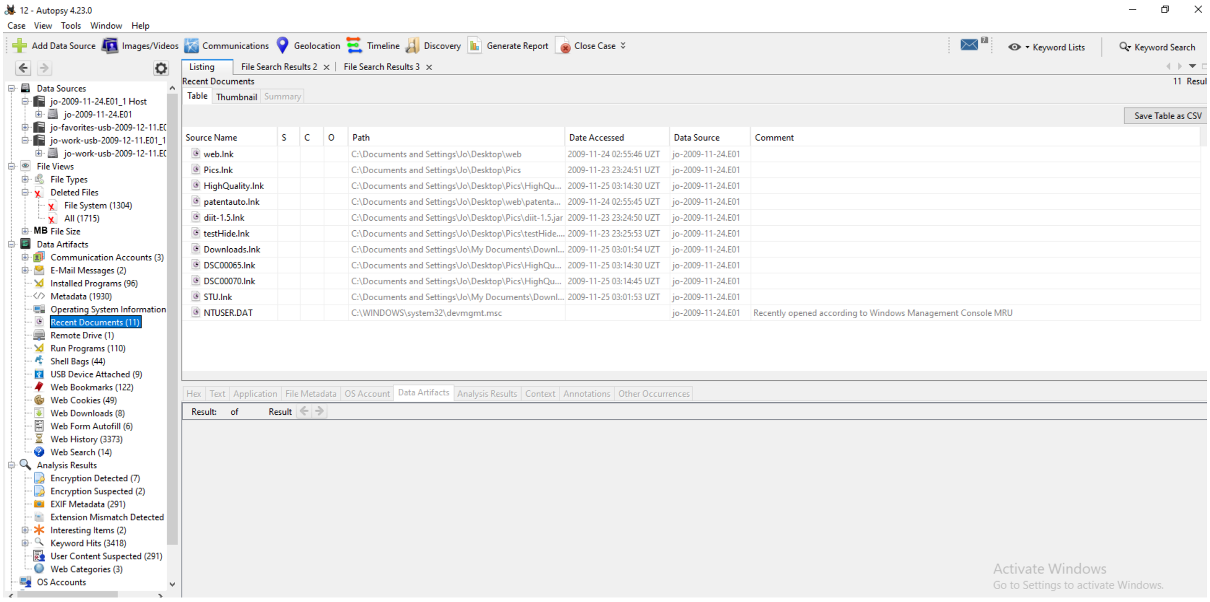
Email Artifacts

Under **Data Artifacts**, click on **E-Mail Messages**.



LNK Files

Under **Data Artifacts**, click on **Recent Documents**.



Intent & Premeditation Analysis

- **Web History & Search:** Analysis of the web history reveals routine, non-malicious browsing (e.g., news, Python documentation, Firefox extensions). There is zero evidence of the user researching competitors or data exfiltration methods, indicating a complete lack of premeditated intent.
- **Email Artifacts:** No unauthorized communications with competitors were found.
- **LNK Files (Recent Documents):** The `Recent Documents` artifact contains only 11 entries. Crucially, there are **no LNK files** corresponding to the 1,600+ stolen PDF documents. This confirms the exfiltration was performed programmatically by malware, bypassing the Windows Shell, rather than through manual user interaction.
- **Root Cause Confirmation:** The presence of `STU.lnk` confirms the user manually executed the disguised malicious payload, falling victim to a phishing/drive-by attack rather than acting as an insider threat.

The Master Event Sequence

Timestamp	Source Artifact	Event Description	Significance
Nov 16, 2009	**RAM Dump**	The malicious macro inside the downloaded document executes. soffice.bin (PID 2392) spawns multiple hidden cmd.exe processes.	**The Compromise:** Marks the exact moment the attacker gained a backdoor into Jo's system via fileless execution.
Nov 23, 2009	**Web History**	A connection is made to python.org, and python-2.6.4.msi is downloaded to the machine.	**Post-Exploitation:** The attacker uses their backdoor to install Python, setting up tools for future data theft.
Nov 25, 2009	**Web History & LNK Files**	Jo manually clicks and accesses the STU.doc file, which was originally hosted on a compromised NOAA .gov website.	**The Vector:** Proves Jo was a victim of a phishing/drive-by attack. The LNK file confirms her manual interaction with the disguise, not a hacking tool.
Dec 10, 2009	**Work USB**	1,632 confidential PDF files (e.g., 10728.MeshNetworks...pdf) are created simultaneously in 17 folders (Papers1 - Papers17).	**The Exfiltration:** The actual data theft. The synchronized timestamps prove an automated script copied the files, not a human.
Dec 10, 2009	**Recent Documents**	**Absence of Evidence:** Zero .lnk files are generated for the 1,600+ PDF files copied to the USB.	**The Alibi:** Mathematically proves Jo did not manually open and copy these files. It was done by the malware in the background.

Post-Dec 10	**Main Disk**	The original PDF files are permanently deleted from Jo's My Documents folder, yielding zero hash matches on the main drive.	**Anti-Forensics:** The attacker wiped the original files to cover their tracks and frame the user for the missing data.

Network C2 Gap:
Because Volatility 3 dropped support for legacy Windows XP network structures, and the malware utilized fileless injection , we could not identify the exact IP address the `cmd.exe` processes were communicating with. This gap requires pivoting to network firewall logs to find the attacker's origin.

The Physical USB Gap:
We know the files were copied to the Work USB on December 10, and we know the USB contains Apple Mac `._` artifacts. However, the endpoint evidence cannot tell us *who* physically plugged that USB in, or if the malware mounted a remote drive to look like a local USB.

Part D — Executive Reporting

On November 16, 2009, the user ("Jo") downloaded a disguised malicious document (`STU.doc`) from a compromised website. This triggered a fileless malware infection. On December 10, 2009, an automated script used this infection to steal 1,632 confidential PDF files by copying them to an external USB drive. The attacker then deleted the original files from the computer to cover their tracks. Forensic evidence proves the user was a victim of a cyberattack, not an insider threat.

Methodology

- **Disk & USB Analysis:** Performed using Autopsy (v4.23.0) to analyze file artifacts, timestamps, and deleted data.
- **Memory Analysis:** Performed using Volatility 3 to analyze active processes and memory injections.

Conclusion The investigation confirms that the data exfiltration was fully automated by malware. The user is entirely cleared of any suspicion regarding the stolen data.